

Semi-Supervised Learning Techniques

1. Simulate the semi-supervised learning techniques using the libraries in Python:
 - a. Consider any image classification dataset (e.g., CIFAR-10 or MNIST). Assume that it has X records.
 - b. Randomly select **100 labelled images**.
 - c. For the **remaining images** (X - 100 records), strip off the labels and treat them as unlabeled data.
 - d. Using the below given semi-supervised learning techniques, generate pseudo-labels for the X - 100 records and compare their accuracy with the ground truth labels.
 - Self-training
 - Co-training
 - Using Clustering: Use K-Means to group similar data. If a cluster contains labeled data, the same label is applied to unlabeled data within that cluster. If a cluster contains multiple labels, you may consider the majority label.

Moodle Submission Logistics:

- You need to have the following files in a zipped folder with the naming format:

“SS_<<roll_no>>”

- (a) Your code files with proper naming extension
- (b) Your dataset/s
- (c) Your output files arranged in order with proper naming extension
- (d) Any reference materials you have used

Reference:

1. https://mvllearn.github.io/references/semi_supervised.html
2. https://scikit-learn.org/stable/modules/generated/sklearn.semi_supervised.SelfTrainingClassifier.html